Advanced ML classification Techniques

Dr. Zainab Mirza

Classification: Basic Concepts

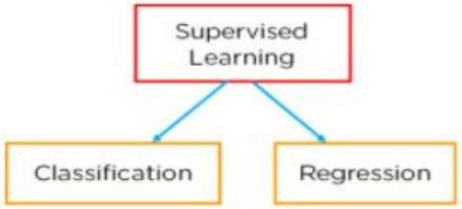
- Classification: Basic Concepts-
- It is a supervised learning technique that is used to identify the category of new observations on the basis of training data

Supervised vs. Unsupervised Learning

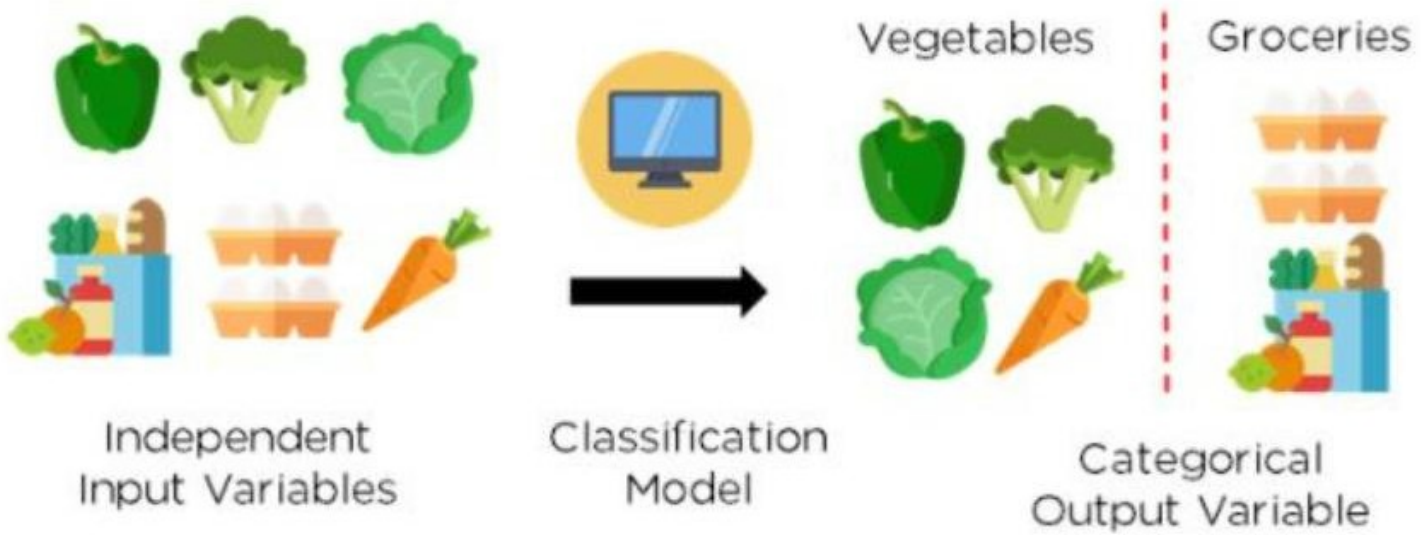
- Supervised learning (e.g. classification)
  - Supervision: The training data (observations, measurements, etc.) are accompanied by **labels** indicating the class of the observations
  - New data is classified based on the training set
- Unsupervised learning (e.g. clustering)
  - The class labels of training data is unknown
    - Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

Prediction Problems: Classification vs. Numeric Prediction

- Classification
  - predicts categorical class labels (discrete or nominal)
  - classifies data (constructs a model) based on the training set and the values (class labels) in a classifying attribute and uses it in classifying new data
- Numeric Prediction
  - models continuous-valued functions, i.e., predicts unknown or missing values
- Typical applications
  - Credit/loan approval:
  - Medical diagnosis: if a tumor is cancerous or benign
  - Fraud detection: if a transaction is fraudulent
  - Web page categorization: which category it is



Classification

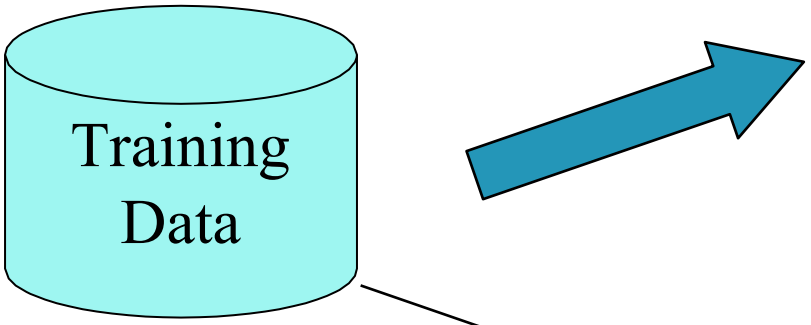


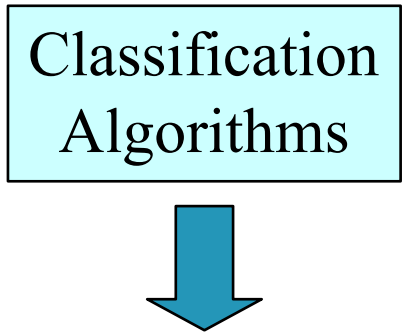
Classification of Vegetables and Groceries

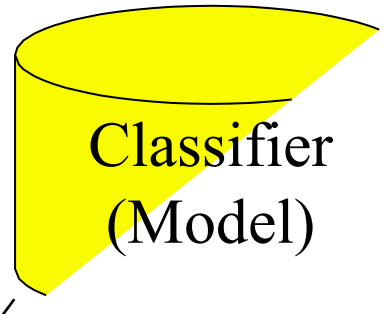
Classification—A Two-Step Process

- Model construction(learning step or training phase):
  - Classifier is built describing a set of predetermined classes
  - Each tuple/sample is assumed to belong to a predefined class, as determined by another database attribute called the class label attribute
  - The set of tuples used for model construction is training set
  - The model is represented as classification rules, decision trees, or mathematical formulae
- Model usage(classification step): for classifying future or unknown objects
  - Estimate accuracy of the model
    - The known label of test sample is compared with the classified result from the model
    - Accuracy rate is the percentage of test set samples that are correctly classified by the model
    - Test set is independent of training set (otherwise overfitting)
  - If the accuracy is acceptable, use the model to classify new data
- Note: If *the test set* is used to select models, it is called validation (test) set

Process (1): Model Construction



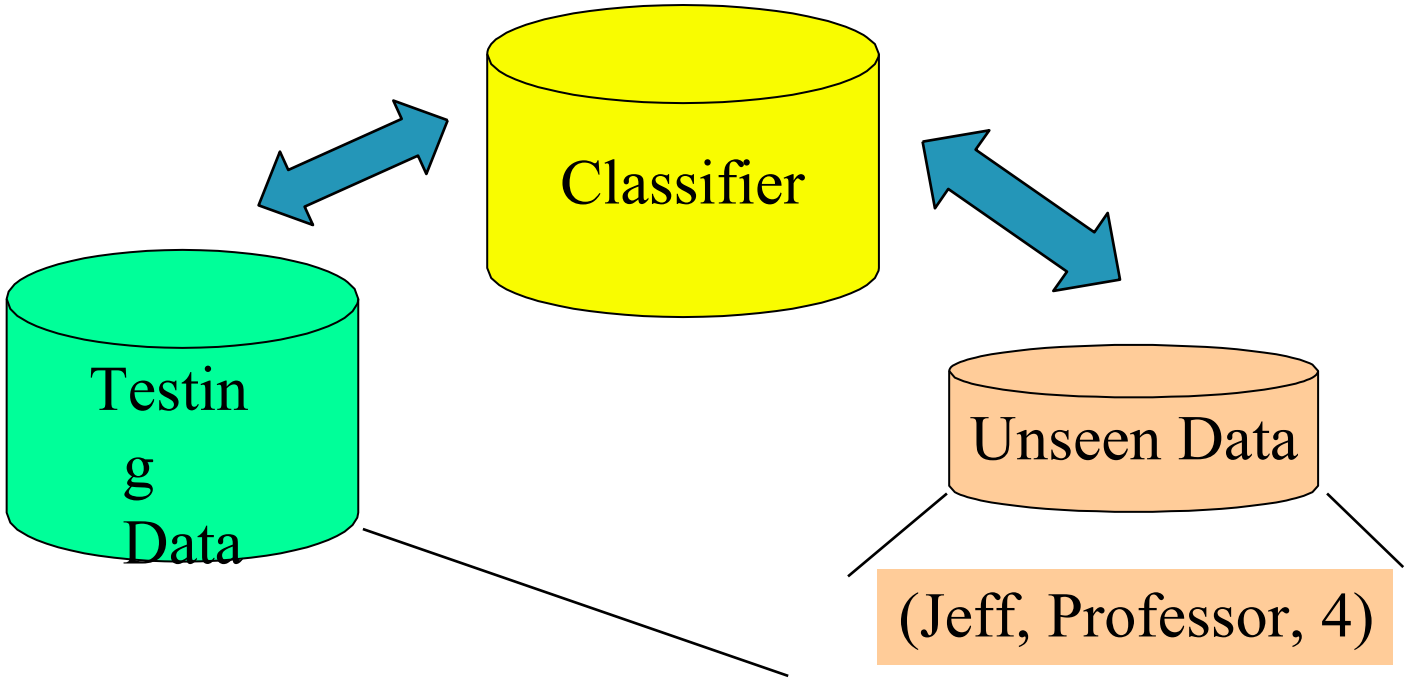




NAM E	RANK	YEARS	TENURED
M ik e	Assistan t Prof	3	no
M ary	Assistan t Prof	7	yes
B ill	Professor	2	yes
J im	Assoc iate Prof	7	yes
Dave	Assistan t Prof	6	no
Anne	Assoc iate Prof	3	no

IF rank = ‘professor’
OR years > 6
THEN tenured = ‘yes’

Process (2): Using the Model in Prediction



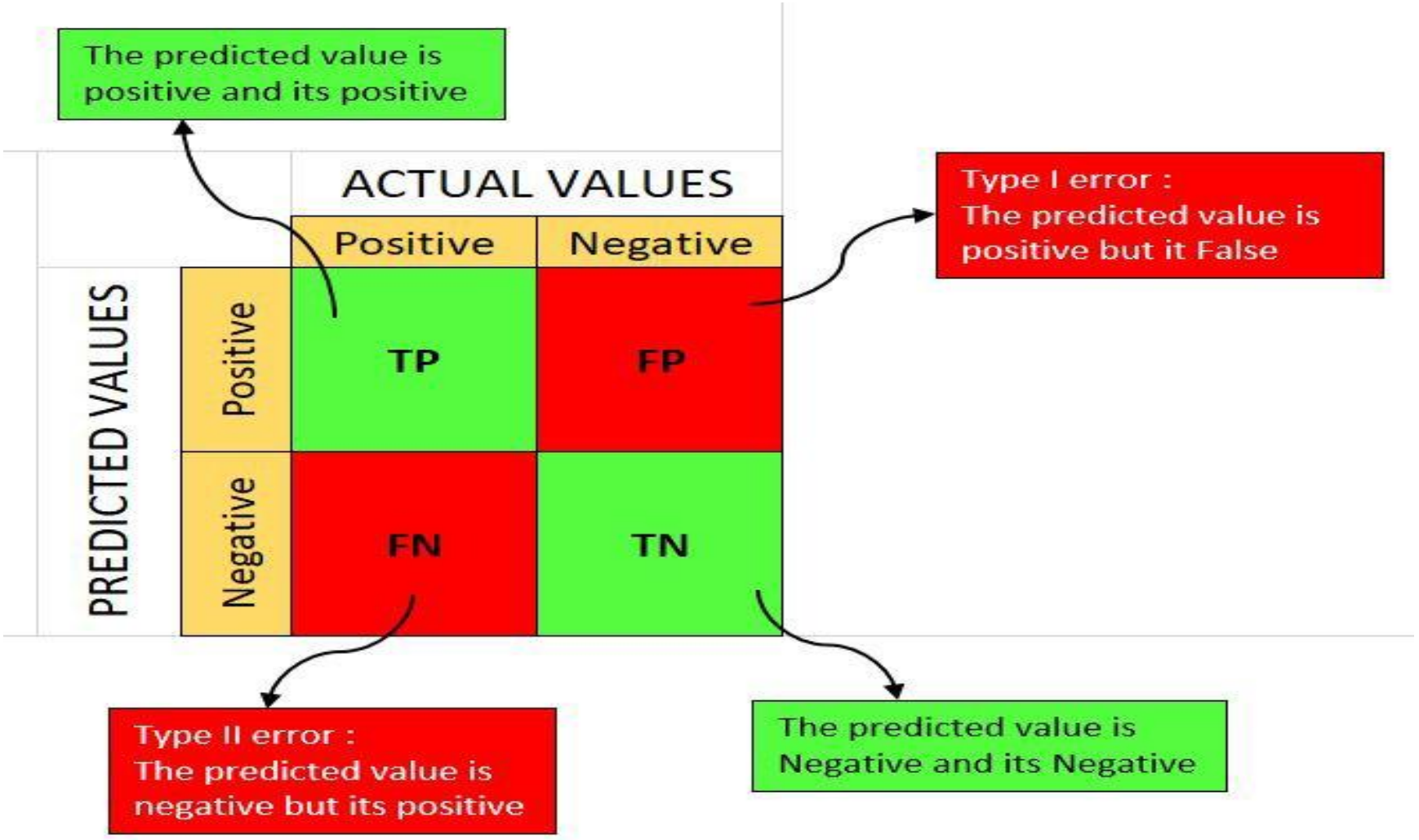
NAM E	RANK	YEARS	TENURED
Tom	Assistan t Prof	2	no
M erlisa	Assoc iate Prof	7	no
G eorge	Professor	5	yes
Joseph	Assistan t Prof	7	yes

Dr.Zainab Mirza
Prof. Anagha

Metrics for evaluating classifier performance

- Accuracy
- Error rate
- Precision
- Sensitivity (Recall)
- Specificity
- F1 Score
- AUC-ROC[Area under curve – Receiver Operator Characteristic]

Confusion Matrix



Confusion matrix- solved example

Y-target =[0,1,0,1,1,0]

Y- predicted =[1,1,0,1,0,1]

Model Evaluation and Selection

Measure	*Formula*
accuracy, recognition rate	$\frac{TP+TN}{P+N}$
error rate, misclassification rate	$\frac{FP+FN}{P+N}$
sensitivity, true positive rate, recall	$\frac{TP}{P}$
specificity, true negative rate	$\frac{TN}{N}$
precision	$\frac{TP}{TP+FP}$
F, F_1, F-score, harmonic mean of precision and recall	$\frac{2 \times precision \times recall}{precision + recall}$
F_β, where β is a non-negative real number	$\frac{(1+\beta^2) \times precision \times recall}{\beta^2 \times precision + recall}$

Evaluation measures. Note that some measures are known by more than one name. TP, TN, FP, P, N refer to the number of true positive, true negative, false positive, positive, and negative samples, respectively (see text).

Model Evaluation and Selection

- **True positives** (*TP*): These refer to the positive tuples that were correctly labeled by the classifier. Let *TP* be the number of true positives.
- **True negatives** (*TN*): These are the negative tuples that were correctly labeled by the classifier. Let *TN* be the number of true negatives.
- **False positives** (*FP*): These are the negative tuples that were incorrectly labeled as positive (e.g., tuples of class *buys_computer* = *no* for which the classifier predicted *buys_computer* = *yes*). Let *FP* be the number of false positives.
- **False negatives** (*FN*): These are the positive tuples that were mislabeled as negative (e.g., tuples of class *buys_computer* = *yes* for which the classifier predicted *buys_computer* = *no*). Let *FN* be the number of false negatives.

Classifier Evaluation Metrics: Confusion Matrix

Confusion Matrix:

Actual class\Predicted class	C_1	$\neg C_1$
C_1	**True Positives (TP)**	**False Negatives (FN)**
$\neg C_1$	**False Positives (FP)**	**True Negatives (TN)**

Example of Confusion Matrix:

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	**6954**	**46**	7000
buy_computer = no	**412**	**2588**	3000
Total	7366	2634	10000

- Given *m* classes, an entry, $\boldsymbol{CM_{i,j}}$ in a **confusion matrix** indicates # of tuples in class *i* that were labeled by the classifier as class *j*

- May have extra rows/columns to provide totals

Classifier Evaluation Metrics: Accuracy, Error Rate, Sensitivity and Specificity

A\P	C	¬C	
C	**TP**	**FN**	**P**
¬C	**FP**	**TN**	**N**
	P'	**N'**	**All**

- **Classifier Accuracy,** or recognition rate: percentage of test set tuples that are correctly classified

  Accuracy = (TP + TN)/All

- **Error rate:** *1 – accuracy*, or

  Error rate = (FP + FN)/All

- **Class Imbalance Problem**:
  - One class may be *rare*, e.g. fraud, or HIV-positive
  - Significant *majority of the negative class* and minority of the positive class
  - **Sensitivity**: True Positive recognition rate
    - **Sensitivity = TP/P**
  - **Specificity**: True Negative recognition rate
    - **Specificity = TN/N**

Classifier Evaluation Metrics: Precision and Recall, and F-measures

- **Precision**: exactness – what % of tuples that the classifier labeled as positive are actually positive

$$precision = \frac{TP}{TP + FP}$$

- **Recall:** completeness – what % of positive tuples did the classifier label as positive?

$$recall = \frac{TP}{TP + FN}$$

- Perfect score is 1.0
- Inverse relationship between precision & recall
- ***F* measure (*F_1* or *F*-score)**: harmonic mean of precision and recall,

$$F = \frac{2 \times precision \times recall}{precision + recall}$$

- ***F_β*:** weighted measure of precision and recall
  - assigns ß tim

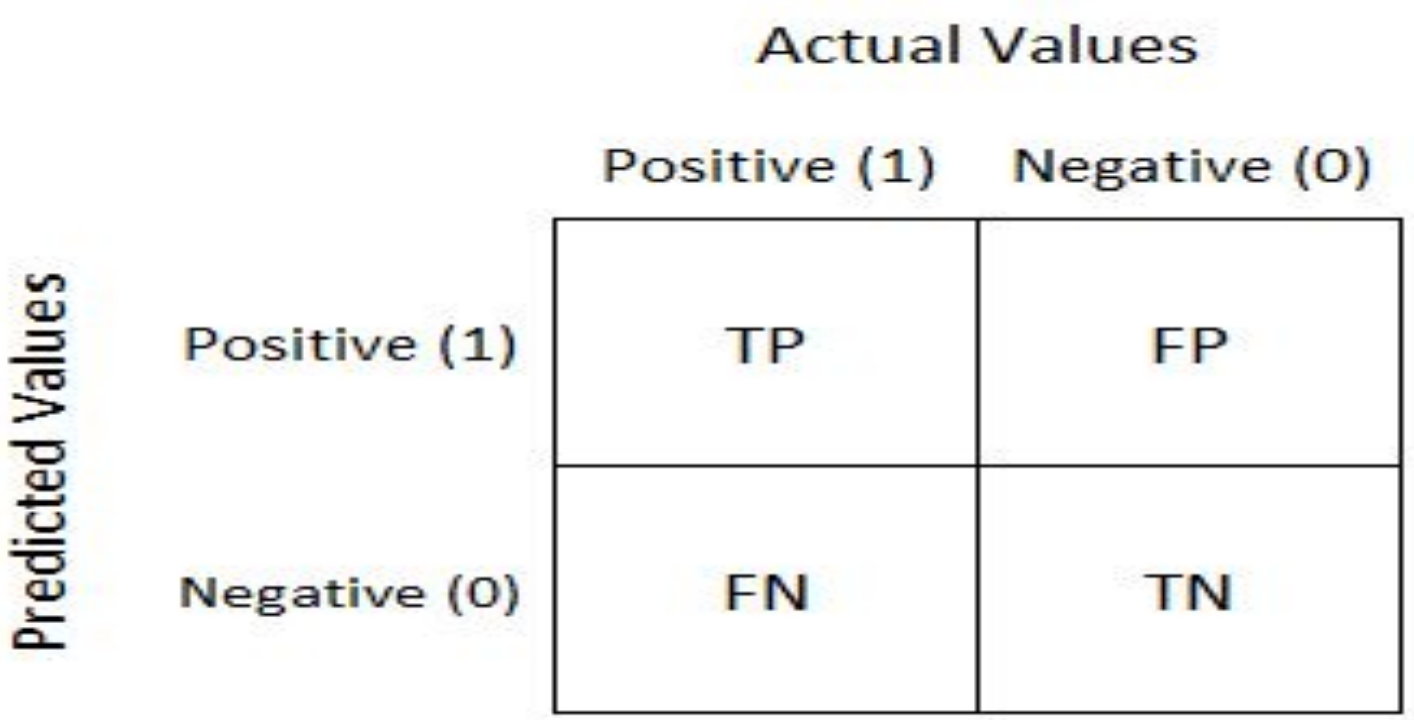
$$F_\beta = \frac{(1 + \beta^2) \times precision \times recall}{\beta^2 \times precision + recall}$$

Classifier Evaluation Metrics: Example

Actual Class\Predicted class	cancer = yes	cancer = no	Total	Recognition(%)
cancer = yes	**90**	**210**	300	30.00 (*sensitivity*
cancer = no	**140**	**9560**	9700	98.56 (*specificity)*
Total	230	9770	10000	96.40 (*accuracy*)

- *Precision* = 90/230 = 39.13% *Recall* = 90/300 = 30.00%

Confusion Matrix



Techniques to improve classification Accuracy

- Bagging [Bootstrap aggregating]
- Boosting

Ensembles: Another Approach

- Instead of training different models on same data, train same model multiple times on different data sets, and "combine" these "different" models
- We can use some simple/weak model as the base model
- How do we get multiple training data sets (in practice, we only have one data set at training time)? One option is to use the ideas illustrated below:



(note: green points are the selected ones)

Figure courtesy: http://quantdare.com/

Ensemble Learning

Bagging: It is a homogeneous weak learners' model that learns from each other independently in parallel and combines them for determining the model average.

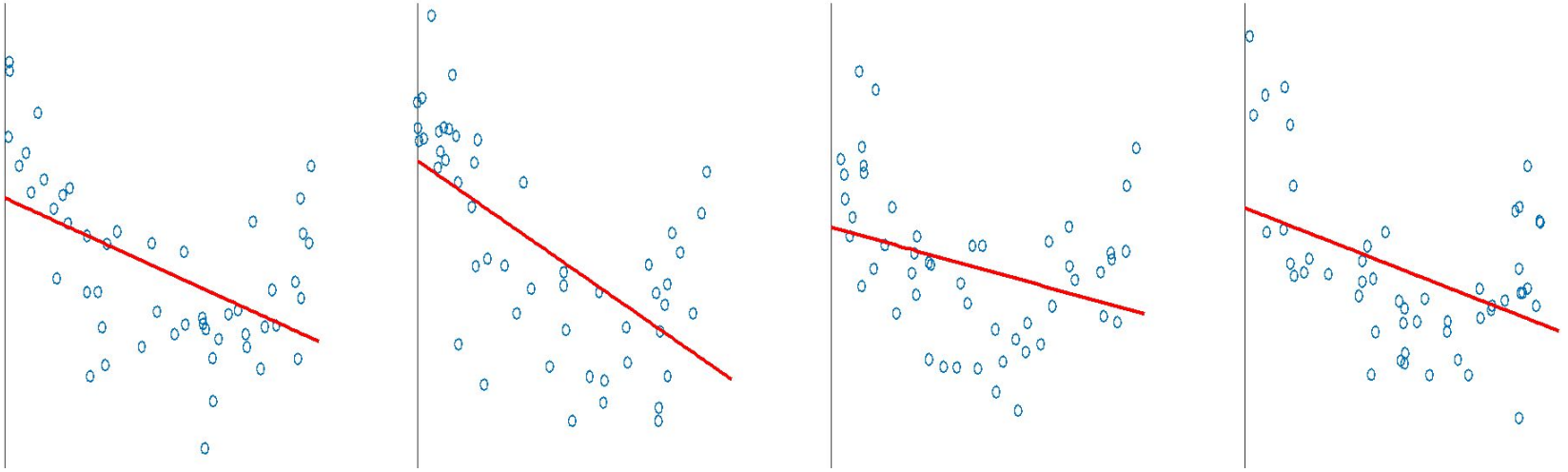
Boosting: It is also a homogeneous weak learners' model but works differently from Bagging. In this model, learners learn sequentially and adaptively to improve model predictions of a learning algorithm.

Bagging

Bootstrap **A**ggregating, also known as bagging, is a machine learning ensemble meta-algorithm designed to improve the stability and accuracy of machine learning algorithms used in statistical classification and regression. It decreases the variance and helps to avoid overfitting. It is usually applied to decision tree methods. Bagging is a special case of the model averaging approach.

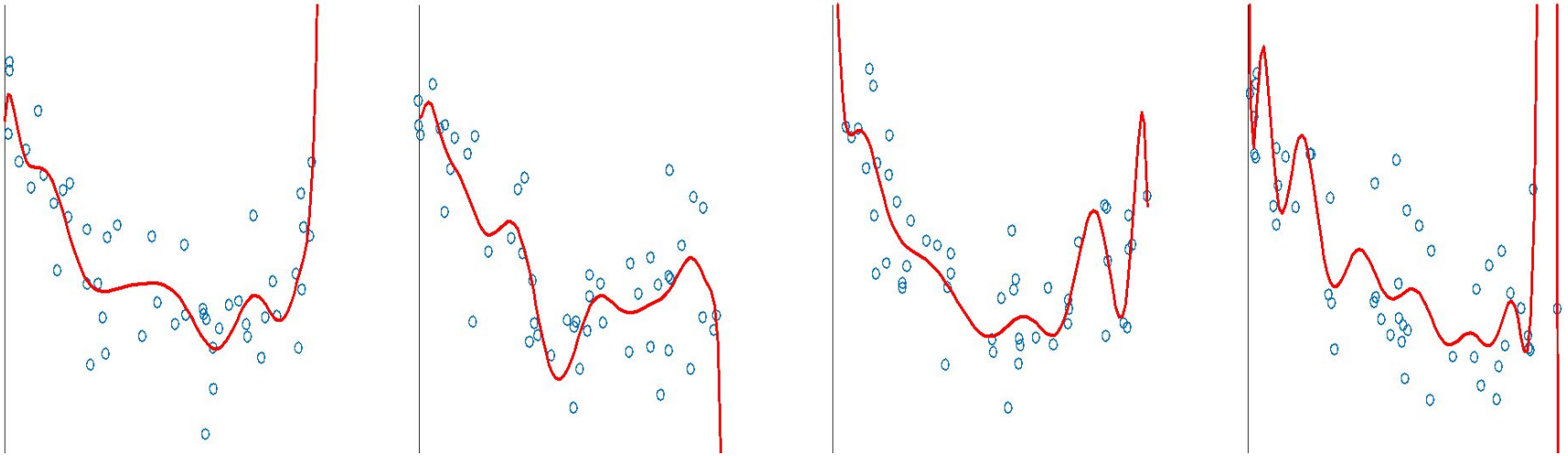
Bias and Variance

Regardless of training sample, or size of training sample, model will produce consistent errors. Bias is one type of error that occurs due to wrong assumptions about data such as assuming data is linear when in reality, data follows a complex function.



Bias and Variance

Different samples of training data yield different model fits. In machine learning variance is the amount by which the performance of a predictive model changes when it is trained on different subsets of the training data.



Bagging

Implementation Steps of Bagging

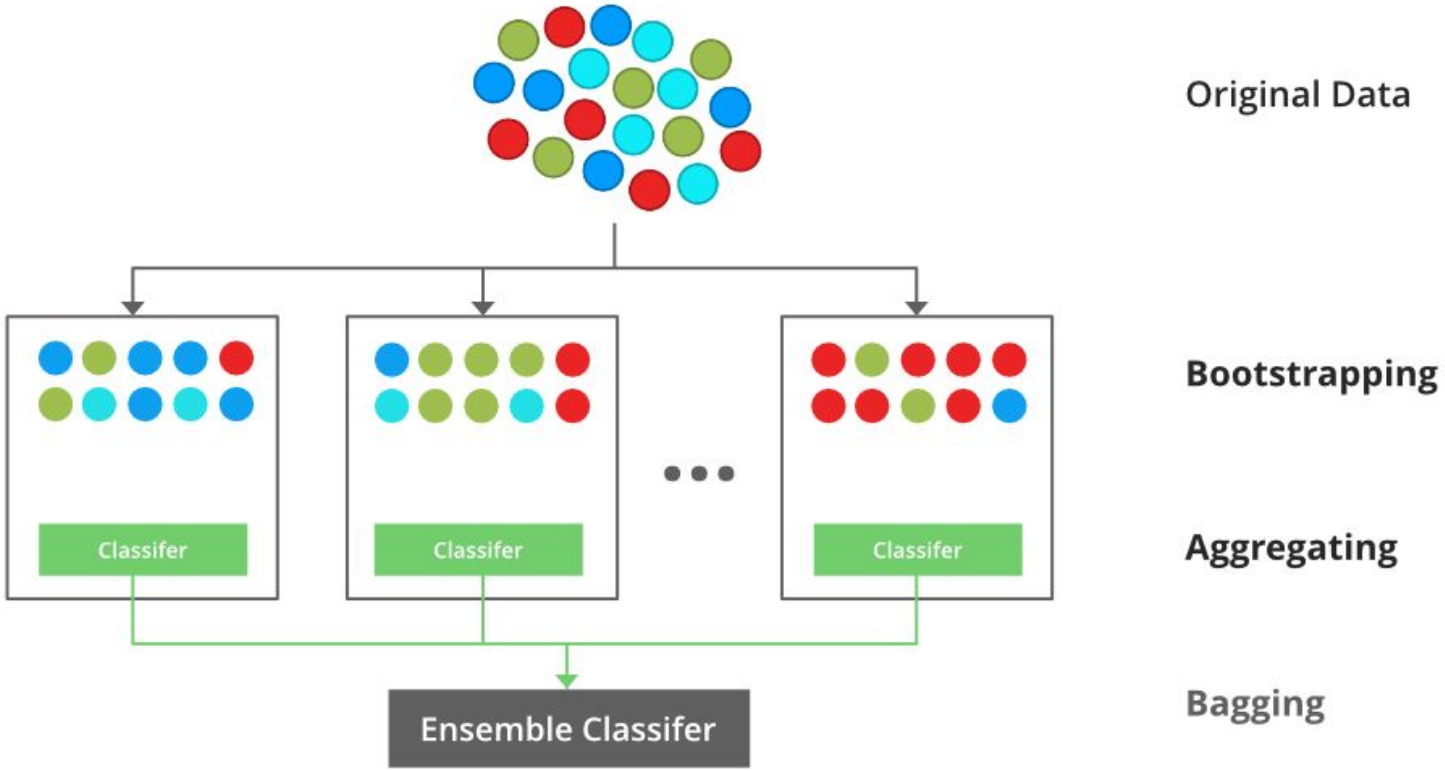
Step 1: Multiple subsets are created from the original data set with equal tuples, selecting observations with replacement.

Step 2: A base model is created on each of these subsets.

Step 3: Each model is learned in parallel with each training set and independent of each other.

Step 4: The final predictions are determined by combining the predictions from all the models.

Bagging


Original Data
Classifer
Classifer
Classifer
Bootstrapping
Aggregating
Ensemble Classifer
Bagging

Boosting

Boosting is an ensemble modeling technique that attempts to build a strong classifier from the number of weak classifiers. It is done by building a model by using weak models in series. Firstly, a model is built from the training data. Then the second model is built which tries to correct the errors present in the first model. This procedure is continued and models are added until either the complete training data set is predicted correctly or the maximum number of models is added.

Boosting

1-Initialise the dataset and assign equal weight to each of the data point.

2- Provide this as input to the model and identify the wrongly classified data points.

3- Increase the weight of the wrongly classified data points and decrease the weights of correctly classified data points. And then normalize the weights of all data points.

4- if (got required results)

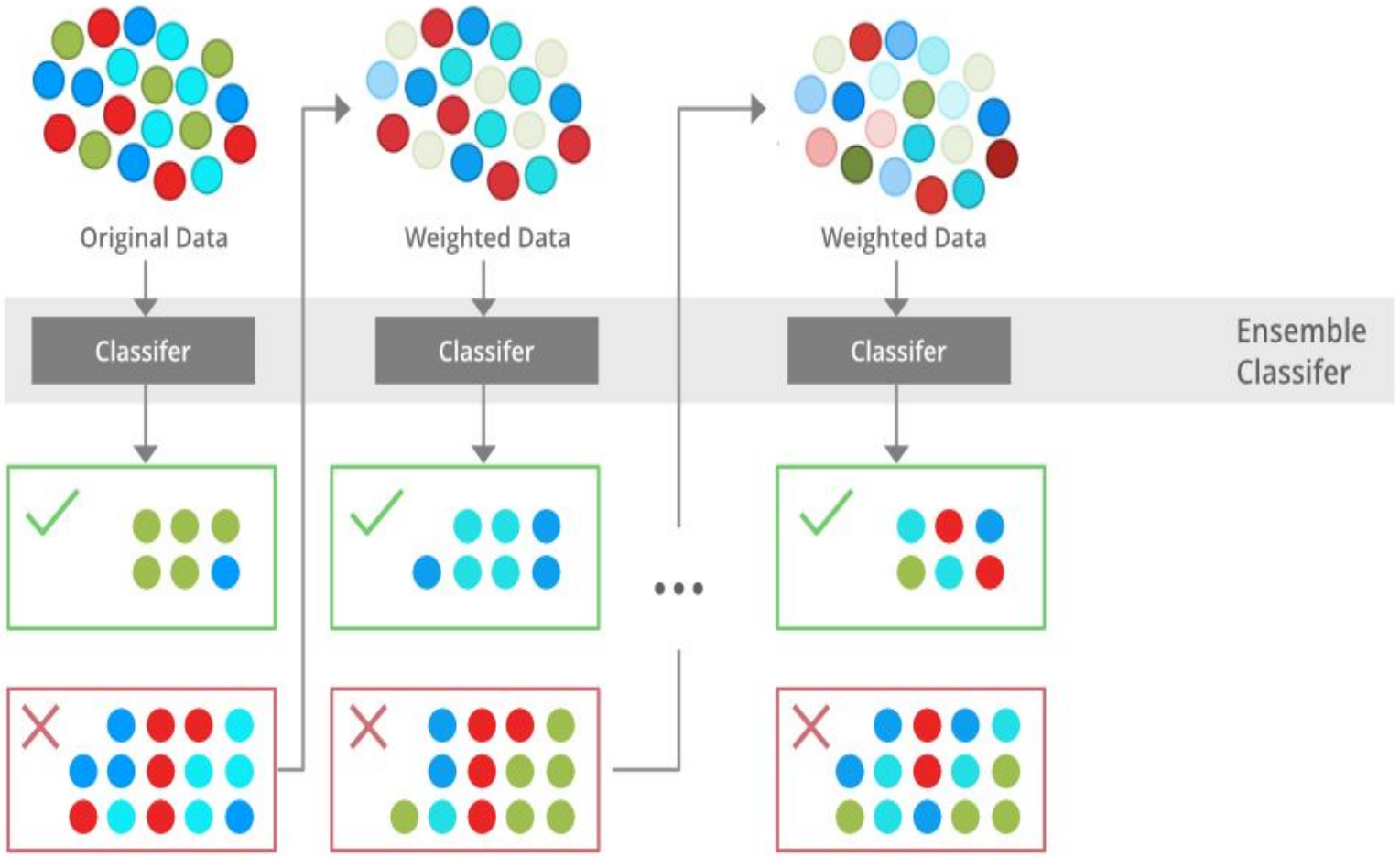
  Goto step 5

else

  Goto step 2

5- End

Boosting


Original Data
Weighted Data
Weighted Data
Classifer
Classifer
Classifer
Ensemble
Classifer

Similarities between Bagging and Boosting

Similarities between Bagging and Boosting are as follows:-

- **Both are ensemble methods to get N learners from 1 learner.**
- **Both generate several training data sets by random sampling.**
- **Both make the final decision by averaging the N learners (or taking the majority of them i.e Majority Voting).**
- **Both are good at reducing variance and provide higher stability.**

Differences between Bagging and Boosting

Differences between Bagging and Boosting are as follows:-

Bagging is the simplest way of combining predictions that belong to the same type while **Boosting** is a way of combining predictions that belong to the different types.

Bagging aims to decrease variance, not bias while **Boosting** aims to decrease bias, not variance.

In **Baggiing** each model receives equal weight whereas in **Boosting** models are weighted according to their performance.

In **Bagging** each model is built independently whereas in **Boosting** new models are influenced by performance of previously built models.

Bagging vs. Boosting

In **Bagging** different training data subsets are randomly drawn with replacement from the entire training dataset.
In **Boosting** every new subsets contains the elements that were misclassified by previous models.
Bagging tries to solve over-fitting problem while **Boosting** tries to reduce bias.
If the classifier is unstable (high variance), then we should apply **Bagging**. If the classifier is stable and simple (high bias) then we should apply **Boosting**.
Bagging is extended to Random forest model while **Boosting** is extended to **Gradient boosting**.

Holdout method and Random subsampling

Holdout Method is the simplest sort of method to evaluate a classifier. In this method, the data set (a collection of data items or examples) is separated into two sets, called the **Training set and Test set**.

Example –

If there are 20 data items present, 12 are placed in training set and remaining 8 are placed in test set.

After partitioning data set into two sets, training set is used to build a model/classifier.

After construction of classifier, we use data items in test set, to test accuracy, error rate and error estimate of model/classifier.

…..Random subsampling

Random subsampling is a variation of the holdout method in which the hold out method is repeated k times . We form ‘K ‘ replica of a given data. For each iteration a fixed number of observation is chosen and it is kept aside as test set.

The overall accuracy estimate is taken as the average of the accuracies obtained from each iteration

Bias / Variance ?

…… is simply defined as the inability of the model because of that there is some difference or error occurring between the model's predicted value and the actual value.

Bias / Variance ?

In machine learning …… is the amount by which the performance of a predictive model changes when it is trained on different subsets of the training data.

Under Fitting / Over fitting

High Bias, Low Variance: A model with high bias and low variance is said to be ……

High Variance, Low Bias: A model with high variance and low bias is said to be ……